AlifBee

Product, Pedagogy and Design Blueprint

A mobile-first Arabic alphabet, reading and writing academy for children and adults. Interface language: French. This document is the single source of truth for the AI builder: read it fully, then build phase by phase.

Version	1.0
Target	Mobile web (PWA), correct on desktop
UI language	French only (Arabic script for content)
Audience	Children (6-12) and adults, switchable mode
Stack suggestion	React + Vite + Tailwind, service-layer data access

0. How the AI builder must use this document

This blueprint is intentionally long and exact. A builder that skims it will produce the generic result the project is trying to escape. Follow these rules.

- **Build in the phases of Section 17.** Never attempt everything in one pass. Finish a phase, run its acceptance checklist, then continue.
- **Do not invent features.** If something is not in this document, do not add it. If something is ambiguous, pick the simplest option and list it under 'Assumptions' in your reply.
- **One idea per screen.** Every screen has exactly one primary action. If a layout needs more than one, split the screen.
- **Wordless first.** Icons, color, motion and sound carry meaning. Text confirms; it never explains alone. A child who cannot read French yet must still succeed.
- **All learner-facing text is French.** Use the copy deck in Section 14 as the tone reference. Code, comments and variable names are English.
- **Anti-patterns in Section 16 are hard bans.** Check them before declaring a phase done.
- **Preserve what works.** The existing tracer with a vector mask that turns the ink red outside the guide, the mastery counter, the daily streak and the bottom navigation are good ideas. Keep the behavior, redesign the presentation.

What exists today and why it must change

Observed problem	Consequence	Fix in this blueprint
Header holds logo, name, dark mode and Eleve/Prof switch; Prof is cropped	Cluttered; a child sees a teacher control	Header = logo + streak only. Role comes from login (Sec. 3, 4)
Bottom nav plus a second scrolling tab strip with cropped labels	Two navigation layers, hidden content	One bottom nav. Sub-views become steps inside a screen (Sec. 3)
Long titles, subtitles and descriptions on every card	Beginners and kids do not read; screens feel heavy	Max one short label per card (Sec. 8)
Level badge says B1 while only 5 of 28 letters are known	Contradictory, CEFR means nothing for an alphabet	Replace with letter-family path and stars (Sec. 5, 6)
Letter grid: index number, check, two buttons per card, order 02 before 01	Letter is not the hero; confusing order	One tap target per letter, correct order, state shown by color and icon (Sec. 4.5)
Features organized as Alphabet / Calligraphie / Harakat tabs	Learner must decide what to do	One guided loop per letter (Sec. 5)
No video showing how a letter is written	Weakest point for a writing-first course	Stroke-order animation plus filmed paper clip (Sec. 7)

1. Product vision and audience

1.1 One-sentence promise

In five minutes a day, a learner sees how a letter is written, hears it, traces it, writes it on paper, and is corrected by a real teacher.

1.2 Why this product is different

- **Writing is the core.** Most apps teach recognition only. This one teaches the hand: stroke order, direction, proportions, dots.
- **A human teacher closes the loop.** The learner photographs their paper work; a teacher marks it correct or not correct with an optional comment. Machine tracing is practice; the teacher is the exam.
- **Progressive by construction.** A learner never sees a letter they have not learned. Every example word, quiz and reading text is generated only from unlocked letters (Sec. 6.4).
- **One app, two temperaments.** Kids mode and Adult mode share content and data but differ in density, tone, motion and reward style (Sec. 9).

1.3 Audiences

Audience	Needs	Design consequence
Children 6-12	Play, big targets, sound, quick wins, parental trust	Giant letter, mascot, stars, minimal text, voice instructions, short sessions (5 min)
Adults	Clarity, speed, explanations of rules, respect for their time	Compact layout, short rule cards, progress overview, no childish rewards
Teacher (20-30 students each)	Fast correction queue, easy content upload	Mobile-friendly correction in under 20 seconds per photo
Admin / owner	Codes, payments, content, certificates	Simple panel, no code needed to change videos or questions

1.4 Success metrics

- Day-1: learner completes the first letter loop in under 6 minutes without help.
- Week-1 retention: at least 40 percent return on day 3.
- Writing submission rate: at least 60 percent of learners who finish a trace also submit a paper photo.
- Teacher turnaround: 90 percent of submissions corrected within 48 hours.

2. Brand and design system

2.1 Emotional tone

Warm, confident, crafted. Think a modern calligraphy studio, not a school worksheet and not a generic SaaS dashboard. Calm negative space, one strong color, ink-like contrast, and a sense of handwriting everywhere.

2.2 Color tokens (CSS variables)

Use exactly these. One dominant brand color (deep zellige green), one sharp action accent (saffron), strict semantic colors. Purple-on-white and beige or cream backgrounds are banned.

```
:root {
  --brand:        #0B6B5D;  /* zellige green, dominant */
  --brand-deep:   #07463D;
  --ink:          #0A1F1B;  /* text and ink strokes */
  --paper:        #EEF3F0;  /* app background, cool mist */
  --surface:      #FFFFFF;
  --accent:       #FFB400;  /* saffron: the ONE primary action color */
  --accent-ink:   #3A2600;  /* text on accent */
  --good:         #1FB57A;  /* correct, mastered */
  --bad:          #FF5A4E;  /* outside the guide, wrong */
  --locked:       #B8C4BF;
  --line:         #C5D1CC;
  --radius-lg: 28px; --radius-md: 18px; --radius-sm: 12px;
}
[data-theme="dark"] {
  --paper:#08201C; --surface:#0E2F29; --ink:#F2FAF7; --line:#1F4A42;
}
```

Rules: saffron is used only for the single primary button of a screen and for the active step. Green is structure. Red appears only for errors and ink leaving the guide. Never use more than three colors in one viewport besides the letter itself.

2.3 Typography

Role	Font	Use	Notes
Display	Gloock (editorial serif)	Screen titles, big numbers	Short titles only, max 3 words, never more than 2 lines
Body / UI	Figtree	Labels, buttons, paragraphs	Forbidden: Inter, Roboto, Arial, Space Grotesk
Mono label	DM Mono	Tiny uppercase tags like STEP 2/5, counters	Used sparingly, 11-12px, letter-spacing 0.08em
Arabic display	Amiri (Naskh), fallback Noto Naskh Arabic	Letters, words, harakat	Always lang='ar' dir='rtl', line-height at least 1.9 so dots and marks never clip
Arabic tracing	SVG paths only	Tracing and animation	Never trace over a font glyph; use authored stroke paths (Sec. 7)

Line length: any paragraph is capped at 60 characters per line on mobile (max-width: 32ch to 36ch) and never exceeds 75 on desktop. Body text 17px minimum on mobile, 1.5 line-height. Type scale: 12 / 14 / 17 / 22 / 30 / 44 / 120 (giant letter).

2.4 Layout language

- **Swiss-grid, asymmetric.** A 4px base unit, 20px screen margins. Left-aligned titles; the giant letter may overlap the card edge by 12-24px to create depth.

- **No boxy card grids.** Avoid identical rectangles in rows. Use one hero element per screen, followed by a vertical path or a single-column list with generous spacing (24-32px between groups).
- **Layers.** Hero card sits on a soft green block that peeks behind it; the letter breaks the card boundary. Use shadows sparingly: one soft shadow style, 0 12px 32px rgba(7,70,61,0.14).
- **Negative space is a feature.** A screen with 40 percent empty space is correct.
- **Shapes.** Large radii (28px) for hero surfaces, 18px for controls, full-round for icon buttons.

2.5 Mobile rules (non-negotiable)

- Viewport: width=device-width, initial-scale=1, viewport-fit=cover. Respect safe-area insets top and bottom.
- Every tap target is at least 48 by 48 px (hard minimum 44). Gap between targets at least 8px.
- No hover dependency anywhere. Every hover state has an equivalent pressed state (scale 0.97, 80ms).
- Thumb zone: primary actions live in the bottom 40 percent of the screen. Sticky bottom action bar for the main button on lesson screens.
- Secondary choices open as **bottom sheets** (drag handle, snap at 50 and 90 percent, backdrop tap closes). Never use centered modal popups.
- Never allow horizontal scrolling of the page. A horizontal carousel is allowed only for the unlocked-letter picker, with visible partial next item.
- Respect prefers-reduced-motion: replace animations with fades.
- Orientation: portrait first. In landscape, tracing canvas goes full screen.

2.6 Iconography

Use one consistent outlined icon set with 2px strokes and rounded caps (Lucide or Phosphor). Every icon appears with a short label in the bottom nav and on first use of a control; afterwards kid mode may show icon only with an accessible aria-label. Required icon vocabulary:

Meaning	Icon idea	Meaning	Icon idea
Learn / path	Route or map pin	Letters	Letter-form glyph
Read	Open book	Progress	Trophy or chart
Watch	Play in circle	Listen	Speaker with waves
Trace	Finger with line	Write on paper	Pencil and sheet
Send photo	Camera	Quiz	Question bubble
Locked	Padlock	Mastered	Filled star
Streak	Flame	Settings	Gear (inside menu only)

2.7 Motion

- Micro-interactions 120-220 ms, ease-out. Page transitions: slide horizontally in the reading direction of French (left to right), but letter animations follow Arabic direction (right to left).
- Success: letter pulses once, a star flies to the counter, soft haptic (navigator.vibrate(15)) and a short chime. Error: the ink turns red, a gentle shake, no negative sound.
- Never auto-play audio on page load. Always play after a tap (browser policy and kindness).

3. Information architecture

3.1 Navigation

Four bottom tabs only. Each tab opens a single-purpose screen. There is no second tab strip anywhere in the learner app.

Tab (French label)	Replaces today	Purpose	Default screen
Parcours	Parcours	The guided path: what to do now	Home with Today card and path
Lettres	Alphabet + Calligraphie + Harakat tabs	Free reference to all unlocked letters; practice any letter	Letter picker (single list)
Lecture	Lecture	Reading practice with only unlocked letters	Reading list by family
Moi	Coaching	Progress, streak, certificates, settings, account	Progress summary

3.2 Header (top bar)

- Left: small AlifBee logo. Right: streak flame with number. That is all.
- Dark mode, language, mode switch (Kids/Adult), logout and activation code live in Moi > Reglages.
- The teacher and admin interfaces are separate routes (/prof, /admin) reachable only after login with a teacher or admin role. They are never shown to learners.

3.3 Route map

```
/                       -> redirect to /parcours
/parcours               Home: Today card + letter-family path
/parcours/famille/:id   Family detail (letters in this family, status)
/lettre/:id             Letter lesson (5-step loop, Sec. 4.3)
/lettre/:id/tracer      Tracing screen (full screen)
/lettre/:id/ecrire      Paper writing + photo submission
/lettre/:id/quiz        Letter quiz
/lettres                Letter picker (all letters, locked ones grey)
/lecture                Reading list
/lecture/:id            Reading exercise
/moi                    Progress, certificates, settings
/moi/reglages           Settings (mode, theme, sound, code)
/acces                  Paywall and activation code
/connexion              Login / sign up
/prof                   Teacher: correction queue
/prof/eleve/:id         Teacher: one student
/prof/contenu           Teacher: videos and questions
/admin                  Admin: codes, students, teachers, certificates
```

3.4 Roles

Role	Can	Cannot
Learner (Eleve)	Learn, trace, submit photos, take quizzes, view own progress	See other learners, content not yet unlocked
Teacher (Prof)	See assigned students, correct submissions, edit own content (videos, questions), comment	Manage codes or payments, see unassigned students
Admin	Everything teacher can, plus codes, assignment of students to teachers, certificates, global content	Nothing restricted

4. Screen-by-screen specification

Each screen lists purpose, content in order from top to bottom, primary action, states, and French copy. The builder implements exactly these.

4.1 Parcours (Home)

Purpose: answer 'what do I do now?' in one second.

- **Top bar:** logo, streak flame + number.
- **Hero card (the Today card):** deep green surface with a soft saffron glow behind. Left: mono tag 'AUJOURD'HUI' and Display title with the letter name ('Baa'). Right: the giant Arabic letter (120px) overlapping the card's top-right edge by 16px. Under it: three small step dots (Voir, Ecouter, Tracer...) showing progress for this letter.
- **Primary button** (saffron, full width, 56px, pinned above the bottom nav): 'Commencer' with play icon. If a lesson is half done: 'Continuer'.
- **Path (below the fold):** a vertical winding path of letter-family nodes (Sec. 6.2). Node states: done (green star), current (saffron, pulsing once), next (outlined), locked (grey padlock). Tapping a done or current node opens the family detail as a bottom sheet.
- **Daily challenge:** a slim row under the hero: flame icon + 'Defi du jour' + 1 line. Completed state: check icon and 'Termine'. Never a second big button.

Empty/first-run: before any progress, the hero says 'Bienvenue' and the letter Alif; no streak displayed.
Copy: keep titles under 3 words.

4.2 Family detail (bottom sheet)

- Title: family letters shown big in a row (e.g., Alif, Baa, Taa, Thaa), each a 56px circle with status ring.
- Tap a letter to open its lesson. Locked letters show a padlock and shake gently when tapped with a small toast 'Bientot !'.
- Footer: progress bar and 'x / n lettres'.

4.3 Letter lesson (the 5-step loop)

This is the heart of the app. It is a single screen with a stepper at the top (5 dots) and a sticky bottom action bar. The content area swaps between steps with a horizontal slide. The learner can go back; progress is saved at every step.

Step	Label (FR)	What the learner does	Content	Done when
1	Voir	Watches how the letter is written	Stroke-order animation (Sec. 7.1), loop, speed toggle 1x / 0.5x, replay button	Animation watched once fully
2	Ecouter	Hears the letter name, sound, and with 3 short vowels	Big speaker button; 4 chips: letter, a, i, u (fatha, kasra, damma). Browser voice (Sec. 11.3)	Each of the 4 chips tapped once
3	Tracer	Traces the letter with a finger on the guided canvas	Tracing canvas (Sec. 7.3) with 3 modes: guide with arrows, guide light, no guide	Passes the check twice (once with guide, once without)
4	Ecrire	Writes it on paper, sends a photo (optional but encouraged)	Filmed clip of hand on paper (Sec. 7.2), downloadable PDF sheet, camera button	Photo sent or 'Passer pour l'instant' tapped

Step	Label (FR)	What the learner does	Content	Done when
5	Retenir	Quick recognition quiz (3 questions)	Sec. 6.5 quiz types, voice read-out of prompts	3 questions answered; at least 2 correct to earn the star

- The letter is always displayed in Arabic font at the top of steps 2 and 5 for continuity. In step 1 and 3 it is the SVG shape.
- The sticky bottom bar has one saffron button whose label changes with the step: 'Suivant', 'Valider', 'Envoyer la photo', 'Terminer'. A secondary icon-only text button 'Passer' appears only on step 4.
- Finishing a letter triggers the celebration (Sec. 2.7) and a bottom sheet: 'Bravo ! Baa est a toi' with two choices: 'Lettre suivante' (primary) and 'Retour au parcours'.
- Mastery: a letter becomes 'maitrisee' (gold star) only after the review in Sec. 6.6 succeeds, not at first completion. First completion gives a green check.

4.4 Tracing screen

- Full screen. Top: close icon, letter name, replay-animation icon. Center: the canvas with a light dotted guide and numbered start dots with arrows (mode 1).
- Bottom: three icon buttons: Effacer, Voir l'animation, Mode (guide / leger / sans guide) and the saffron 'Valider'.
- Live feedback while drawing: ink is dark green inside the guide corridor and turns red when outside; releasing the finger evaluates the stroke immediately and shows a soft green flash or a gentle shake.
- Validation algorithm in Sec. 7.4. Kid mode accepts any stroke order; Adult mode can enforce order (setting).

4.5 Lettres (letter picker)

- Single-column vertical list in the correct alphabet order (Alif first). Each row: 64px Arabic letter on the right side (RTL feel), name on the left, and a state icon at the far edge: padlock, check, gold star.
- A segmented control at the top toggles the view: 'Liste' / 'Grille'. Grid view uses 4 columns of 72px circles (letter only, state by ring color). Default for Kids: grid; for Adults: list.
- Tapping an unlocked letter opens a bottom sheet with 4 big icon buttons: Voir, Ecouter, Tracer, Ecrire. This replaces the two buttons on every card today.
- Order must be strictly Alif, Baa, Taa, ... Never show 02 before 01.

4.6 Paper writing and photo submission

- Top: the filmed clip (Sec. 7.2) in a 16:9 frame, plus a link 'Telecharger la feuille' (PDF, Sec. 7.5).
- Middle: three illustrated tips as icon rows: 'Ecris 5 fois', 'Lignes droites', 'Bonne lumiere'.
- Bottom: saffron button 'Prendre une photo' (opens camera input, accept=image/*, capture=environment), secondary 'Choisir une photo'.
- After capture: preview with crop/rotate, button 'Envoyer'. Compress client-side to max 1600px long edge, JPEG 0.8, under 800 KB.
- Status chip after sending: 'En attente' (grey), 'Correct' (green), 'A refaire' (red) with the teacher's optional comment shown as a speech bubble. A 'Reessayer' button re-opens the camera.

4.7 Quiz screen

- One question per screen, giant prompt, 2-4 large answer buttons (min 64px high), progress dots on top.

- Voice read-out of the prompt via speaker icon. Kid mode reads it automatically after a tap on the first question.
- Feedback: immediate green or red state with a one-line French reason; a 'Continuer' button appears. No timers in v1.
- Result screen: stars (1-3), 'Refaire' and 'Continuer'. The pass threshold for exams is 75 percent (Sec. 6.5).

4.8 Lecture

- List of reading exercises grouped by unlocked families, each a single row: title (Arabic word), small icon (word type) and state.
- Exercise screen: the word in large Arabic text with letters color-highlighted on tap (reveals the letter name and sound); button 'Ecouter'; 'J'ai lu' marks completion.
- Words are drawn only from unlocked letters (Sec. 6.4). Provide the transliteration toggle 'Aa' (default off for adults, on for kids).

4.9 Moi (progress, settings)

- Top: avatar (initial) and name, streak flame, total stars.
- Progress ring: letters mastered out of 28, with the 7 families as segments of the ring.
- Certificates: appears once all letters are mastered and the final exam is passed (name printed, Sec. 13.4).
- Reglages (list rows with icons): Mode (Enfant / Adulte), Theme (clair / sombre), Sons, Voix, Code d'activation, Aide (WhatsApp), Deconnexion.

4.10 Acces (paywall and activation)

- The first family (Alif, Baa, Taa, Thaa) is free so the learner can feel the product before paying.
- Locked families show a padlock. Tapping one opens a bottom sheet: title 'Debloquer tout le cours', three payment option rows with icons (PayPal, WhatsApp, Carte) opening the external payment links, and a text field 'J'ai un code' with a saffron 'Activer'.
- Activation: code validated against the database; success unlocks permanently; error gives a clear French message. See Sec. 13.
- Never block the free family or the settings.

4.11 Connexion

- Email + password, or magic link. Role chosen by the admin, not the user. Kid accounts can be created by a parent: parent email, child first name, birth year (no exact date).
- One screen, one button 'Continuer'. Show a short privacy line and a link to the policy.

5. Pedagogy

5.1 Principles

- **See, hear, move, write, recall.** Each modality reinforces the others. The letter loop (Sec. 4.3) always follows this order.
- **Small steps, mastered.** One letter per session, 3 to 5 minutes. A learner who wants more can continue, but the default pace is one letter per day.
- **Shape families.** Letters that share a skeleton are taught together (ba, ta, tha) so the learner focuses on the single difference: the dots.
- **Dots are meaning.** Dots get their own mini-step inside tracing: after the body is traced, place the dots (count and position).
- **Sound attached early.** The letter is never taught as a bare shape. Short vowels (fatha, kasra, damma) are heard in step 2 from the first letter.
- **Recall beats completion.** A star is earned by remembering after a delay, not by finishing a lesson.
- **Human feedback for the hand.** The photo correction by a teacher is the authority for writing quality.
- **Errors are information.** No scolding. The app shows what to fix: 'Les points sont en dessous'.

5.2 The learning loop per letter

```
Voir (animation) -> Ecouter (name + a/i/u) -> Tracer (guided, then unguided)
   -> Ecrire (paper + photo) -> Retenir (3 questions)
        |
        v
   Review scheduled: +1 day, +3 days, +7 days, +21 days (Sec. 6.6)
```

5.3 Session design

- Target session: 5 minutes. Hard stop suggestion after 15 minutes: 'Belle seance ! Reviens demain'.
- Kids: maximum 1 new letter per day by default, optional bonus review.
- Adults: may unlock the next letter immediately after mastery or after the delay chosen in Reglages ('Rythme': Calme / Normal / Intensif).

5.4 Forms of the letter

Each letter has up to four forms: isolated, initial, medial, final. The curriculum introduces the isolated form first. Initial/medial/final forms are introduced in a second cycle called 'Liaison' per family, after the family is mastered in isolated form. Six letters never connect to the following letter: alif, daal, dhaal, raa, zaay, waaw; teach this rule explicitly with an icon showing a broken link.

5.5 Harakat and reading rules (cycle 2)

- Short vowels: fatha, kasra, damma. Tanwin. Sukun. Shadda. Introduced after the first 3 families are mastered, always through sound before name.
- Long vowels: alif, waaw, yaa as vowel carriers, introduced with minimal pairs.
- Sun and moon letters (cycle 3), taa marbuta, hamza placements: only after the alphabet is complete.

5.6 Kids vs Adults pedagogy

Dimension	Kids	Adults
Explanations	Voice and animation, no text rules	One short rule card per step, expandable
Rewards	Stars, mascot reactions, stickers	Progress ring, streak, certificates

Dimension	Kids	Adults
Pace default	1 letter per day	Chosen rhythm, 1-3 letters per day
Tracing strictness	Forgiving corridor, any order	Narrower corridor, order enforced optional
Transliteration	On	Off, toggle available
Language	Very short French sentences, tutoiement	Short, neutral French, vouvoiement

6. Curriculum and content model

6.1 The 28 letters

Names use French-friendly transliteration. Codes are Unicode code points for the isolated form (render with the Arabic font; do not retype). 'Connects' = connects to the following letter.

#	Name	Code point	Connects
1	Alif	U+0627	No
2	Baa	U+0628	Yes
3	Taa	U+062A	Yes
4	Thaa	U+062B	Yes
5	Jiim	U+062C	Yes
6	Haa (soft)	U+062D	Yes
7	Khaa	U+062E	Yes
8	Daal	U+062F	No
9	Dhaal	U+0630	No
10	Raa	U+0631	No
11	Zaay	U+0632	No
12	Siin	U+0633	Yes
13	Shiin	U+0634	Yes
14	Saad	U+0635	Yes
15	Daad	U+0636	Yes
16	Taa (emphatic)	U+0637	Yes
17	Dhaa (emphatic)	U+0638	Yes
18	Ayn	U+0639	Yes
19	Ghayn	U+063A	Yes
20	Faa	U+0641	Yes
21	Qaaf	U+0642	Yes
22	Kaaf	U+0643	Yes
23	Laam	U+0644	Yes
24	Miim	U+0645	Yes
25	Nuun	U+0646	Yes
26	Haa (breath)	U+0647	Yes
27	Waaw	U+0648	No
28	Yaa	U+064A	Yes

Display order in the app is always this order. The builder must render letters from these code points and must not rely on the screenshot examples.

6.2 Letter families (the path)

Seven families. Each family is a node on the path. Recommended pace: one family per week for adults, 4 to 5 days for kids (one letter a day plus review).

Family	Letters	Teaching focus	Free?
F1 Racines simples	Alif, Baa, Taa, Thaa	Same body, dot differences; sound of first vowels	Free
F2 Crochets	Jiim, Haa, Khaa	Hook shape; dot position changes meaning	Paid
F3 Sans liaison	Daal, Dhaal, Raa, Zaay	Letters that do not connect; broken-link rule	Paid
F4 Dents et ventres	Siin, Shiin, Saad, Daad	Teeth and bowl shapes; emphatic contrast	Paid
F5 Lettres fortes	Taa (emphatic), Dhaa (emphatic), Ayn, Ghayn	Emphatic and throat sounds; listening	Paid
F6 Boucles	Faa, Qaaf, Kaaf, Laam	Loops and tall stems; Laam with Alif	Paid
F7 Rondes et finales	Miim, Nuun, Haa (breath), Waaw, Yaa	Final shapes; vowel carriers	Paid

6.3 Content record per letter

```
Letter {
  id: "baa", order: 2, family: "F1",
  nameFr: "Baa", codePoint: "U+0628",
  connects: true,
  forms: { isolated, initial, medial, final },   // SVG stroke data (Sec. 7)
  soundUrl: null,                // v1 uses browser voice, see 11.3
  speechText: "baa",             // text passed to speech synthesis
  vowels: { a: "ba", i: "bi", u: "bu" },
  dots: { count: 1, position: "below" },
  similarTo: ["taa", "thaa", "nuun", "yaa"],  // for quiz distractors
  videoStrokeAnimationId: "baa_isolated",
  videoPaperUrl: "https://youtube.com/...",   // private/unlisted link, editable by teacher
  tip: "Le point est sous la lettre."          // one line only
}
```

6.4 Progressive vocabulary rule

Every word, example and reading text carries a list of required letters. The app shows an item only if all its required letters are in the learner's mastered or current set. Content authors enter words once; filtering is automatic. If the list for a family is empty, the app shows the letter-only practice. Never display a letter or word the learner cannot yet decode.

6.5 Quiz types and rules

Type	Prompt	Answers	Used for
Letter from sound	Speaker plays the name	4 letter buttons, distractors from similarTo	Recognition
Sound from letter	Big letter shown	3 audio buttons	Mapping
Dot placement	Letter body without dots	Tap where dots go	Dot discipline
Form matching	Isolated form	Pick initial/final form from 3	Cycle 2 liaison
Odd one out	4 letters, 3 share a family	Tap the different one	Differentiation
Vowel read	Letter with fatha/kasra/damma	3 audio buttons	Harakat

- Letter quiz: 3 questions, star rating 1-3 (2 correct = 1 star, 3 correct = 3 stars).
- Family exam: 10 questions drawn from the family's letters; **pass = 75 percent**. Failure unlocks nothing and offers 'Revoir' with the missed letters.
- Final exam: 25 questions across all families; pass = 75 percent; success unlocks the certificate.
- Question bank must be editable in the teacher panel (Sec. 12). The builder seeds at least 6 questions per letter and 15 per family.

6.6 Review scheduling (simple spaced repetition)

```
Box 1: review after 1 day      Box 2: after 3 days
Box 3: after 7 days            Box 4: after 21 days   Box 5: mastered (gold star)
Correct answer -> move up one box. Wrong answer -> back to Box 1.
Daily review queue = letters whose due date <= today, max 5 per day.
Mastered = reached Box 4 and answered correctly once more.
```

6.7 Points, stars and streak

- Points exist but are secondary. 10 per step, 30 per star-quiz, 50 for mastery. Show points only in Moi.
- Streak: counts days with at least one finished step. A single 'repair day' per week is allowed without losing the streak.
- Never punish with negative scores. Never show leaderboards for children.

7. Writing videos, tracing and paper practice

This chapter specifies the feature the owner cares about most: showing exactly how each letter is written.

7.1 Stroke-order animation (primary)

Each letter form is authored as an SVG made of ordered strokes. The same data drives three things: the animation (Voir), the guided tracing (Tracer) and the validation mask (check). This guarantees what the learner watches is exactly what they trace.

- Animation draws each stroke with the stroke-dashoffset technique, in order, with a small pen-tip dot leading the stroke and a brief pause between strokes.
- Strokes are drawn right-to-left where the script requires it; dots appear last with a small pop.
- Controls: replay, speed (1x and 0.5x), 'pas a pas' step button (shows stroke 1, then 2...). Loop automatically after a 1.2 s pause.
- Numbered start dots appear before each stroke begins to indicate where to start.
- Weight: under 4 KB per form; no video file needed; plays offline.

Stroke data schema

```
GlyphForm {
  letterId: "baa", form: "isolated", viewBox: "0 0 200 200",
  baseline: 120,                       // y of the writing line
  strokes: [
    { id: 1, kind: "body",
      d: "M 170 80 C 150 110, 100 130, 40 105",   // authored path
      width: 14, corridor: 22,                    // draw width, tolerance
      startDot: [170, 80], direction: "rtl" },
    { id: 2, kind: "dot", at: [100, 150], radius: 7 }
  ],
  dotRule: { count: 1, position: "below" }
}
```

Authoring note: the builder cannot reliably invent correct calligraphic paths for 28 letters x 4 forms. Seed all 28 isolated forms with best-effort Naskh paths, mark each as 'draft' in the data, and create an internal admin screen to preview and edit paths visually (drag control points). The teacher validates each form before it goes live.

7.2 Filmed paper clip (secondary)

A short real-hand video shows writing on paper with a pen or reed, matching what learners do in step 4.

Spec	Value
Duration	20-30 seconds per letter
Framing	Top-down, hand and paper only, no face, 16:9, 1080p, steady tripod
Content	1) Full letter once at normal speed, 2) once slow, 3) the finished letter with a green highlight on the dots
Writing line	Paper has a clear baseline; a faint line drawn so proportions are visible
Audio	No recorded audio. Optional on-screen captions in French (3 words)
Hosting	Private or unlisted YouTube link per letter, stored in the letter record; teacher can replace it from the panel
Embedding	youtube-nocookie.com embed, playsinline, no related videos, controls on, fallback poster image
Naming	baa_isolated_paper.mp4 for master files

7.3 Tracing canvas

- HTML canvas sized to the viewport square, device-pixel-ratio aware, touch-action: none to prevent scrolling while drawing.
- Use pointer events (pointerdown, pointermove, pointerup) with coalesced points for smooth strokes; pressure ignored.
- Three modes: (1) Guide with arrows and start dots, (2) light guide (20 percent opacity), (3) no guide. Passing mode 3 is required for the star in the loop.
- Palm rejection: ignore pointers whose contact width exceeds 40 px if pointerType is touch.
- Rendering: the learner's ink is smoothed with quadratic curves; ink thickness 10-14 px; color var(--brand-deep) inside the corridor and var(--bad) outside.

7.4 Validation algorithm

```
Inputs: learner points P (per stroke), target stroke path S, corridor width w.
1. Sample S at 64 equal-length points.
2. coverage = % of S points having a learner point within w px.
3. outside  = % of learner points farther than w from any point of S.
4. direction (optional): compare start point; allowed distance 2*w.
5. dots: each learner dot tap must be within 2*w of a target dot; count must match.
Pass body stroke if coverage >= 80 percent AND outside <= 15 percent.
Kid mode: coverage >= 70 percent, outside <= 25 percent, order ignored.
Return reason codes: 'trop_court', 'sort_du_guide', 'mauvais_sens', 'points_manquants'
and map them to short French messages (Sec. 14).
```

7.5 Printable writing sheets

- Per letter: one A4 PDF with the letter's four forms on top, a dotted tracing line, then empty guided lines with a baseline. Title 'Feuille d'ecriture - Baa'.
- Generated by the app (client-side) from the stroke data, or uploaded as static PDFs by the teacher. Offer a download button in step 4.
- Include a small QR code or URL footer with the lesson link.

7.6 Photo correction

- Learner sends one photo per letter attempt. The teacher sees: photo, letter reference sheet, and two buttons 'Correct' / 'A refaire' with an optional text comment (max 140 characters) and quick reasons (chips: points, direction, taille, lignes).
- A learner can retry unlimited times; the history of attempts is kept.
- Notification (in-app badge, optional email) to the learner when corrected.

8. Component library

Build these once and reuse. Each is described by anatomy, sizes and states.

Component	Spec
PrimaryButton	Saffron background, accent-ink text, height 56px, radius 18px, full width in sticky bar; pressed scale 0.97; disabled 40 percent opacity; loading spinner replaces icon
SecondaryButton	Transparent, 2px brand border, same size; used at most once per screen
IconButton	48x48, circular, outlined icon 24px, aria-label required
BottomNav	4 items, 72px tall plus safe area, active item has brand-deep pill with white icon and label, inactive icon + label grey; labels never truncated
TodayCard	Hero card, brand-deep background, radius 28, giant letter overlapping, stepper dots, glow
PathNode	72px circle with state ring, connected by a curved line; done = good star, current = saffron, locked = grey padlock
LetterChip	56px circle containing the Arabic letter, status ring; used in sheets and pickers
Stepper	5 dots, current is a saffron pill, past are green, future are outline
BottomSheet	Drag handle, rounded 28px top, backdrop 40 percent ink, snap points 50 and 90 percent, escape on backdrop tap and swipe down
Toast	Top-center, 2.5 s, icon + 1 short line, never blocks input
Mascot (kids)	A bee mascot (AlifBee) in 5 poses: wave, cheer, think, oops, sleep; vector, under 6 KB each
StarRating	1-3 stars, fills with a small bounce
ProgressRing	Segmented 7-part ring with center number
StatusChip	Pill: En attente (grey), Correct (green), A refaire (red)

8.1 Spacing and sizing scale

```
space: 4 8 12 16 20 24 32 40 56 72
screen padding: 20      section gap: 32       card inner padding: 20-24
min touch: 48           bottom nav: 72 + safe-area
max content width on desktop: 480px centered phone-style column, with
a calm side area showing the mascot or empty space (no sidebars).
```

8.2 Desktop behavior

Correct but secondary. The learner app stays in a centered 480px column with the same bottom navigation. The teacher panel is the only place that uses a wide layout (two-column: queue list and photo viewer).

9. Kids mode and Adult mode

A single switch in Moi > Reglages (and chosen at signup) sets mode. The mode is a presentation layer: same data, same lessons.

Element	Kids mode	Adult mode
Type scale	+2px body, titles larger, giant letter 160px	Base scale, giant letter 120px
Layout density	One element per screen block, spacing 32+	Moderate; shows rule card and stats
Navigation labels	Icon + 1 word	Icon + 1 word
Mascot	Present in every reaction	Hidden
Reward	Star fly-in, sticker book, sound chime	Subtle check, progress ring
Quiz	Auto read-out, 3 options max	4 options, optional hint
Letter picker default	Grid of circles	List with names
Transliteration	On	Off
Copy tone	Tutoiement, exclamations ok	Vouvoiement, calm
Parental area	Hidden behind a simple gate ('Appuie 3 secondes')	Not applicable
Theme	Brighter accents allowed (same tokens)	Standard

10. Data model

Implement behind a data-service module (DataService with methods such as getLetters(), saveProgress(), submitPhoto()) so storage can move from local prototype to a real backend without UI changes. If the builder environment offers a database, use it; otherwise start with local storage and keep the interface identical.

```
User        { id, role: learner|teacher|admin, email, displayName, mode: kids|adult,
              birthYear?, teacherId?, createdAt, entitlement: free|full }
Progress    { userId, letterId, stepsDone: [voir, ecouter, tracer, ecrire, retenir],
              box: 0..5, nextReviewAt, stars: 0..3, firstDoneAt, masteredAt? }
Attempt     { id, userId, letterId, type: trace|quiz, score, passed, reason?, createdAt }
Submission  { id, userId, letterId, imageRef, status: pending|correct|redo,
              comment?, correctedBy?, correctedAt?, createdAt }
Letter      { see Sec. 6.3 }
Word        { id, textAr, translationFr, requiredLetters: [letterId], audioText }
Question    { id, letterId?, familyId?, type, prompt, options[], correctIndex, explanationFr }
ActivationCode { code, createdBy, usedBy?, usedAt?, expiresAt?, plan }
Certificate { id, userId, issuedAt, serial }
Streak      { userId, current, best, lastActiveDate, repairUsedWeek }
Setting     { userId, theme, sound, rhythm, voiceName }
```

10.1 Rules

- A learner's unlocked set = F1 always + all families if entitlement is full + families passed in sequence (a family unlocks after the previous family exam is passed or its letters are mastered).
- Photos are stored in private object storage, accessible only by the learner, their teacher and admins; URLs are signed and expire.
- Deleting a learner removes their photos and progress.

11. Technical architecture

11.1 Stack

- React with TypeScript, Vite, Tailwind CSS, React Router. State: lightweight store (Zustand) plus the DataService.
- PWA: manifest, service worker caching shell and stroke data, installable on Android and iOS home screen, offline lesson viewing (animations and tracing work offline; photo uploads queue until online).
- Fonts: self-host Gloock, Figtree, DM Mono, Amiri with font-display: swap and preload for Amiri and Gloock.

11.2 Folder structure

```
src/
  app/            routes, layout, providers
  features/
    parcours/  lettre/  tracer/  ecrire/  quiz/  lettres/  lecture/  moi/  acces/
    prof/  admin/
  components/     buttons, nav, sheets, cards, mascot
  data/           letters.ts, words.ts, questions.ts, glyphs/ (stroke JSON)
  services/       dataService.ts, speech.ts, entitlement.ts, review.ts
  lib/            strokeMath.ts (coverage/outside), image.ts (compress), pdf.ts
  styles/         tokens.css, base.css
```

11.3 Audio and speech

- No recorded audio in v1. Use the browser SpeechSynthesis API with lang 'ar-SA' for letters, words and vowels, and 'fr-FR' for French feedback.
- On first launch, detect available voices; if no Arabic voice exists show a clear fallback: the letter name in French transliteration and a banner 'Installe une voix arabe dans les reglages du telephone' with a help link.
- All speech starts only after a user tap. Provide a global mute in Reglages. Keep a speech.ts wrapper so recorded audio files can replace it per letter later (soundUrl field).

11.4 Performance targets

- First contentful paint under 2 s on a mid-range Android on 4G; initial JS under 200 KB gzipped (lazy load tracer, quiz, teacher panel).
- Tracing runs at 60 fps; no layout shift during animations.
- Images compressed client-side before upload (Sec. 4.6).
- Lighthouse mobile: Performance 85+, Accessibility 95+.

11.5 Accessibility

- Contrast AA for all text; the letter itself at AAA.
- Every icon button has an aria-label in French; focus rings visible (2px saffron).
- Screen-reader labels for tracing: describe the instruction, offer a 'Marquer comme fait' alternative for users who cannot trace.
- Respect reduced motion and text scaling to 200 percent without breaking layouts.

11.6 Privacy and children

- Collect minimum data. Children: first name and birth year only, parent email for the account.
- No advertising, no third-party trackers. Analytics only first-party and anonymous.

- Photos of handwriting may include hands only: instruct 'Photographie seulement la feuille'. Retention: 12 months then auto-delete.
- Provide a privacy page in French and a way to request deletion. Follow GDPR principles and local data-protection law.

12. Teacher and admin panels

12.1 Teacher: correction queue (/prof)

- Mobile-friendly. Top: filters (En attente, A refaire, Tous) as a segmented control. List rows: student first name, letter, time ago, tiny thumbnail.
- Tap a row to open the viewer: large photo (pinch zoom), the reference letter on the side or above, big buttons 'Correct' and 'A refaire', reason chips, optional comment, 'Suivant' to jump to the next pending photo.
- Target: a correction takes under 20 seconds. Keyboard shortcuts on desktop: C, R, N.
- Counters: pending count badge on the nav.

12.2 Teacher: students (/prof/eleve/:id)

- Overview: mastered ring, last activity, streak, recent submissions, quiz scores.
- Message button opening WhatsApp with a pre-filled French message.

12.3 Teacher: content (/prof/contenu)

- Letters list. For each letter: field 'Lien video papier' (paste YouTube link, validates it), 'Astuce' (one line), 'Feuille PDF' upload, and 'Previsualiser'.
- Stroke editor (admin or advanced teacher): path preview, drag nodes, mark validated.
- Questions: list per letter and family, add/edit/delete, type picker, answers with image/letter support, preview in learner style.
- Words: add word, enter required letters automatically detected from the text.

12.4 Admin (/admin)

- Codes: generate a batch (count, plan, expiry), see used/unused, revoke, export CSV.
- Students: search, assign teacher, change entitlement, reset progress, delete.
- Teachers: invite by email, set capacity (20-30 students recommended), see load.
- Certificates: view issued, regenerate, change template text.
- Settings: WhatsApp number, payment links, support email.

13. Access, payment and certificate

13.1 Model

- Free: family F1 (4 letters) with full loop except photo correction limited to 1 per day.
- Paid: all families, unlimited corrections, certificate. One-time payment at launch; subscription kept as an option in the data model (plan field) but disabled.
- Payment happens outside the app (PayPal, WhatsApp, card link). After payment the admin sends an activation code.

13.2 Activation flow

```
Learner taps a locked family -> sheet 'Debloquer tout le cours'
  -> chooses payment method (opens external link) OR enters a code
  -> app validates code: exists, unused, not expired
  -> success: entitlement = full, code marked used, confetti, family opens
  -> failure: 'Ce code n'est pas valide' (no detail about why)
Rate limit: 5 failed attempts per hour per account.
```

13.3 Pricing display

Pricing text and links are configured in the admin settings; the app never hardcodes amounts. Show one clear price and the three payment icons.

13.4 Certificate

- Awarded after all 28 letters are mastered and the final exam is passed at 75 percent or more.
- PDF A4 landscape: AlifBee logo, 'Certificat de reussite', learner name (as typed in profile), date, serial number, teacher/director signature line, QR to verify.
- Name is editable once before issuing; downloaded from Moi.

14. Copy deck (French)

Tone: short, warm, concrete. Kids: tutoiement. Adults: vouvoiement. Titles max 3 words. Buttons are verbs. No exclamation overload.

Key	Kids	Adults
home.hero.tag	AUJOURD'HUI	AUJOURD'HUI
home.cta.start	Commencer	Commencer
home.cta.continue	Continuer	Reprendre
home.challenge	Defi du jour	Defi du jour
step.see	Voir	Observer
step.listen	Ecouter	Ecouter
step.trace	Tracer	Tracer
step.write	Ecrire	Ecrire sur papier
step.remember	Retenir	Retenir
trace.ok	Bravo !	Tres bien.
trace.outside	Reste dans le chemin	Le trace sort du guide.
trace.short	Va jusqu'au bout	Le trace est incomplet.
trace.dots	Les points sont ici	Verifiez la position des points.
quiz.right	Oui !	Correct.
quiz.wrong	Presque. Ecoute encore	Incorrect. Reecoutez la lettre.
photo.send	Envoyer la photo	Envoyer la photo
photo.pending	En attente	En attente de correction
photo.correct	Correct	Correct
photo.redo	A refaire	A refaire
lock.toast	Bientot !	Cette famille est verrouillee.
empty.review	Rien a revoir aujourd'hui	Aucune revision aujourd'hui.
streak.repair	Jour de repos utilise	Jour de repos utilise
offline	Pas de connexion	Vous etes hors ligne.

15. Analytics and quality

- Events (anonymous): lesson_started, step_completed, trace_pass, trace_fail(reason), quiz_completed, photo_sent, photo_corrected, code_redeemed, paywall_viewed, session_length.
- Dashboards for the admin: funnel from first open to first letter done to first photo sent to purchase; top failing letters; average correction time.
- Testing: unit tests for strokeMath and the review scheduler; end-to-end test for the 5-step loop; manual test on 3 real phones (small Android, large Android, iPhone).

15.1 Definition of done for any screen

- One primary action only. Fits in one viewport without scrolling for the main task (path and lists excepted).
- No text block longer than 2 short lines in kids mode; none longer than 4 in adult mode.
- All touch targets at least 48px. No hover-only interaction.
- Works in dark mode and with 200 percent text size.
- All copy in French from the deck; no English leaks.
- Tested at 360 x 740 and 390 x 844.

16. Anti-patterns (hard bans)

Do NOT	Instead
Use Inter, Roboto, Arial or Space Grotesk	Gloock, Figtree, DM Mono, Amiri
Purple gradients, beige or cream backgrounds, generic SaaS look	Zellige green, saffron, cool mist, ink
Grids of identical rounded cards	Hero card, path, single-column lists, deliberate overlap
Two navigation layers or scrolling tab strips	One bottom nav with 4 tabs
Role switch or teacher controls in the learner header	Role-based routes after login
Centered modal popups	Bottom sheets
Hover states as the only feedback	Pressed state, haptics, sound
CEFR level names for the alphabet course	Letter families and stars
Long subtitles and descriptions on cards	One short label
Teaching bare letter shapes without sound	Sound and vowels from step 2
Showing letters or words not yet learned	Progressive vocabulary filter (Sec. 6.4)
Tracing over a font glyph	Authored stroke paths
Auto-playing audio on load	Audio after a tap
Negative scoring, leaderboards for children	Stars, gentle retries
Putting everything on the first screen	One idea per screen

17. Build plan in phases

Phase 1 - Foundation and design system

- Project setup, tokens, fonts, dark theme, safe-area handling, BottomNav, Header (logo + streak), BottomSheet, buttons, toasts.
- Data service interface with local implementation and the letters dataset (Sec. 6.1) with families.
- **Acceptance:** empty app shell with 4 tabs works on a 360px phone; tokens used everywhere; no hardcoded colors; Lighthouse accessibility 95+.

Phase 2 - Parcours and letter loop (without tracing)

- Today card, path, family sheet, letter lesson with steps 1 (animation placeholder), 2 (speech), 5 (quiz), progress saving.
- Review scheduler (Sec. 6.6) and daily queue.
- **Acceptance:** a new user finishes letter 1 in under 6 minutes; progress persists; locked letters behave correctly.

Phase 3 - Stroke animation and tracing

- Stroke data for 28 isolated forms (draft), animation component, tracing canvas, validation algorithm, 3 modes, kid/adult thresholds.
- Internal stroke preview/edit page.
- **Acceptance:** animation and tracing share the same data; red ink outside the corridor; tracing passes at 80 percent coverage; works at 60 fps.

Phase 4 - Paper writing and teacher correction

- Camera capture, compression, upload, status chips, printable sheets, teacher queue and viewer.
- **Acceptance:** learner sends a photo; teacher corrects in under 20 s; learner sees result and retries.

Phase 5 - Letters, reading, Moi

- Letter picker (list/grid), reading exercises with progressive vocabulary filter, progress ring, settings, kids/adult mode switch.
- **Acceptance:** mode switch changes density and tone; no unlearned letters appear anywhere.

Phase 6 - Access, certificate, admin

- Paywall sheet, activation codes with rate limiting, admin code generator, entitlement logic, certificate PDF.
- **Acceptance:** F1 is free, other families lock and unlock via a valid code; certificate shows the learner's name.

Phase 7 - Cycle 2 content and polish

- Liaison forms (initial, medial, final), harakat lessons, family exams, final exam, mascot reactions, PWA offline, performance pass.
- **Acceptance:** Lighthouse targets met; all items in Sec. 15.1 verified on three phones.

18. Ready-to-paste prompts for the AI builder

Paste these in order. Attach this PDF to the first one.

Prompt 1 - Kickoff

```
You are a senior product designer and frontend engineer. Read the attached
blueprint completely. Do not code yet. Reply with: (1) a 10-line summary of
the product, (2) the list of assumptions you will make, (3) any ambiguity you
need resolved. Then wait.
```

Prompt 2 - Phase 1

```
Build Phase 1 exactly as in Section 17 of the blueprint: tokens, fonts,
header, bottom nav (4 tabs), bottom sheet, buttons, dark theme, data service
with the 28 letters and 7 families. French UI. Mobile-first at 360px.
Respect Section 2 and Section 16. When done, list how each acceptance item
was met. Do not start Phase 2.
```

Prompt 3 - Phase 2

```
Build Phase 2: the Parcours home (Today card, path, family sheet) and the
5-step letter lesson (Voir placeholder, Ecouter with browser voice, Retenir
quiz), plus the review scheduler. Follow Sections 4.1 to 4.3, 5 and 6.
Keep every screen to one primary action.
```

Prompt 4 - Phase 3

```
Build Phase 3: stroke-order animation and the tracing canvas using the
GlyphForm schema (Section 7.1) and validation (Section 7.4). Seed draft
strokes for all 28 isolated letters and add the internal stroke preview page.
One data source must drive animation, tracing and validation.
```

Prompt 5 - Phases 4 to 7

```
Continue with Phase N from Section 17, one phase per message. After each
phase, run the checklist in Section 15.1 and report failures honestly.
```

Prompt for redesign review (any time)

```
Audit the current build against Sections 2, 4, 15.1 and 16 of the blueprint.
Give: PURPOSE of the screen, GAP ANALYSIS (what is basic or non-compliant),
ELEVATION PLAN (typography, hierarchy, spacing). Then apply the fixes.
```

19. Open decisions and assumptions

Decision	Default assumed	Owner
Final brand name (AlifBee vs AlifBee Academie)	AlifBee in header, full name on certificate	Owner
Subscription vs one-time payment	One-time; plan field ready	Owner
Free content size	Family F1 only	Owner
Recorded audio vs browser voice	Browser voice v1, soundUrl field ready	Owner
Hosting for photos and data	Backend provided by builder environment, abstracted	Builder
Mascot design	Bee, flat vector, brand colors	Owner approval
Teachers per learner	1 teacher per learner, 20-30 learners per teacher	Owner
Languages beyond French	None in v1; copy externalized to a French file for later translation	Owner
Launch date	Unknown	Owner

Final note to the builder

The product succeeds if a six-year-old and a sixty-year-old can both open it, understand what to do in one second, and finish their first letter without reading a paragraph. When a design choice is in doubt, remove something.